

Research Paper

Using a location-based game to collect preference information for urban and rural forest planning

Philip Chambers^{a,b,*}, Tuulikki Halla^a, Harri Silvennoinen^a, Teppo Hujala^a, Jukka Tikkanen^a^a School of Forest Sciences, University of Eastern Finland, Joensuu, Finland^b Gamification Group, Faculty of Information Technology and Communication Sciences, Tampere University, Finland

HIGHLIGHTS

- Collecting human-centred data for forest planning is resource-intensive and costly.
- Location-based games like geocaching can be utilised in data collection.
- Geocaches closer to urban areas have more visits and higher survey response rates.
- Setting up shorter, accessible, circular trails improves data collection.
- Geocaches collect the most responses in the first summer after establishment.

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ABSTRACT

Location-based games provide an opportunity to collect crowdsourced, spatially explicit data that can be used in decision-making processes in forest planning. While these games have been more widely used in collecting mapping and modelling data, there is limited research on their suitability for collecting location-based data on human preferences, opinions, and emotions in these contexts. Collecting human-centred data can be resource-intensive and costly, hence exploring the potential of location-based games is important. This study investigates the use of geocaching in gathering meaningful data in urban and rural settings. A methodology was developed wherein surveys were embedded in geocaches on trails located in urban and rural forests in Finland. These surveys collected quantitative, qualitative, and photographic data relating to human-forest relationships and landscape preferences. The findings indicate that trails closer to urban areas with higher population density showed increased cache log volume and survey response rates. Moreover, trails tended to collect most information in the first summer after establishment. It is recommended that future studies employing this methodology create shorter, easily accessible, circular trails and that survey questions should be clear and incorporate a method to verify that submitted responses reflect experiences at the chosen location. Furthermore, forest planning professionals and researchers seeking to utilise location-based games in data collection are advised to engage with the communities active in playing them to ensure success.

1. Introduction

Location-based games and gamified fitness applications provide new opportunities to crowdsource spatial data (Heipke, 2010). They can gather detailed place-based information in large quantities, which can aid decision-making. However, there are differences in how useful this data is in urban and rural contexts, as is evident in the scaling and accuracy of mapping and 3-D spatial modelling (Laato & Tregel, 2023). Additionally, the data these games collect is often limited to the physical

characteristics of a place and points of interest chosen by game designers and player communities (Laato et al., 2019), rather than information and locations needed by land-use planners. Landscape planning processes, particularly forest planning, require complex and multi-faceted knowledge and information, including the diverse perspectives of individuals, communities, and user groups, which can vary at different landscape levels. The information needed for urban and peri-urban forests can differ from those required for rural areas. These easily accessible greenspaces tend to have higher visitor numbers and host a broader and

* Corresponding author at: University of Eastern Finland, School of Forest Sciences, P. O. Box 111, FI-80101 Joensuu, Finland.

E-mail address: philip.chambers@uef.fi (P. Chambers).

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evolving range of activities. The increasing popularity of location-based games has led to the emergence of new recreational communities that landscape and forest planners need to understand and engage with. These motivational information systems have changed how people interact with forests and move in nature (Laato et al., 2023). Fuelled by their individual forest relationships, these visitors may attach important meanings to forests, making forest planning and decision-making complex and challenging (Ranacher et al., 2017).

Addressing diverse opinions and feelings in forest planning is a process that requires a thoughtful and inclusive approach, which should consider concepts such as human-forest relationships, landscape preferences, and psychological ownership (Matilainen et al., 2017). Assessing public opinion through participative planning can be costly, time-consuming, and resource-intensive (Mann & Jeanneaux, 2009). Reaching a consensus in these processes can be hindered by participants' interest and knowledge levels, who may also experience fatigue and difficulty when visualising abstract solutions to problems (Roux et al., 2021). Furthermore, these factors may also affect forest planning professionals' decision-making (Predmore et al., 2011). To overcome these obstacles, it is possible to use both active methods, such as Public Participation GIS (PPGIS) (Kytä et al., 2023), and passive methods, such as data mining (Muñoz et al., 2020; Pelka, 2018). These methods, used at different scales and combined with others, can capture the public's thoughts, feelings, and opinions about forests more comprehensively and democratically (Thoneick, 2021). Currently, location-based games and gamified fitness apps don't actively collect users' nuanced and diverse perspectives, so surveys and expert opinions are also needed (Korpilo et al., 2018).

Several factors can affect how people perceive and feel about managed forests, including their attractiveness, peacefulness, biodiversity values, recreation, and income potential (Dearden, 1984). These aspects can be studied using both quantitative and qualitative methods. Location-based games and the communities that play them offer an opportunity to investigate these. This study, utilising the location-based game geocaching, guided players to specific areas within publicly and privately managed forests to record their preferences, thoughts and feelings. Surveys were integrated into the geocaching game, allowing participants to complete them while discovering geocaches along pre-determined routes through different forest environments in urban and rural settings in Finland. The geocaching trail format allowed participants to engage with these topics on-site, providing a novel and contemporaneous experience.

The main objective of this study was to assess whether a location-based game could be used to collect human-forest relationship and landscape preference data on forests and forest management for participatory forest planning processes (RQ1) and to understand this location-based game community better (RQ2). This is achieved by evaluating survey response rates and data quality through statistical analysis, conventional content analysis and photo-coding. The results of this research may have implications for using location-based games as a tool for survey research in landscape planning and management.

2. Background and related work

2.1. Challenges in human-centred data in forest planning

Forest planning involves analysing the social, environmental, and economic objectives of a particular forest area. The process should allow for informed decisions to be made in the development of management strategies which facilitate operations while minimising impact and conflict (França et al., 2022). While the management of stakeholders' diverse views and the avoidance of conflict is a central role of modern forest planning (Mann & Jeanneaux, 2009), active participative approaches at the local level, in both public and private forests, have been thought to be too costly and laborious to adopt (Leskinen, 2004). This has led to a lack of stakeholder engagement and opinion gathering,

which can, in turn, lead to tension and conflict, especially in *peri*-urban and urban areas (Nousiainen & Mola-Yudego, 2022).

In Finland, forests cover approximately 86 % of the total land area, and due to the Everyman's Rights, they are all accessible to the general public and considered multiple-use forests. Finnish forest planning processes have taken two distinct forms. The first form applies to publicly owned and managed forests where participatory methods have been adopted and used in recent decades. This category includes urban forests under municipal management, national parks and other protected areas, which the state forestry agency, Metsähallitus, manages. The second form pertains to privately owned forests and is administered by the forest centre, Metsäkeskus. In this context, participatory processes are not implemented except in regional land use planning (Tikkanen et al., 2010). While some aspects of landscape preferences and human-forest relationships may be considered at the strategic planning level for public and private forests, outside of urban areas, they have not been formally considered at the tactical and operational levels (Weintraub & Cholaky, 1991). With around 96 % of Finns participating in outdoor activities, including nationally popular hobbies such as berry and mushroom picking and outdoor sports such as cross-country skiing (Finnish Environment Ministry, 2022), the potential exists for low-level conflict (Meriläinen & Lehtinen, 2022). This may be compounded by Finland's relatively high ease of access to outdoor recreation opportunities and the culture of owning second 'summer' homes (Paracchini et al., 2014). It is, therefore, important to develop new, accessible methods capable of gathering opinions, preferences, and emotional data in forests and assessing their applicability in forest planning processes.

2.2. Human-forest relationships

By studying human-forest relationships, it is possible to identify, describe and understand the diversity of different, and sometimes contradictory, meanings that individuals and communities attach to forests. These meanings can be deeply linked to their identity and play a significant role in their forest-related attitudes. Exploring these human-forest relationships can contribute to understanding forest-related conflicts (Halla et al., 2023).

Several interview- and survey-based research methodologies exist to explore human meaning-making processes and their human-forest relationships (Apajalahti et al., 2022). However, these can sometimes be time-consuming, involve only a few participants, or may be based more on memorised forest experiences rather than on a lived, onsite experience of the forest (Häggström, 2019; Skär, 2010). As geocaching takes players physically into forest environments, it offers a unique opportunity to collect data on how they experience forests in real-time.

2.3. Landscape preference studies

Managed forests are expected to fulfil diverse economic, environmental, and social roles, including recreation, nature tourism and aesthetic perception (Tyrväinen et al., 2003). Therefore, planning professionals need to consider how landscapes are perceived and enjoyed. Previous research has highlighted the incorporation of landscape preferences in various stages of forest operations (Gundersen & Frivold, 2008). However, distinctions exist between forest planners and forest machine operators, with financial implications dominating the former's concerns and terrain influencing the latter's choices (Karjalainen et al., 2011; Silvennoinen et al., 2011). Landscape issues are often overlooked in discussions between forest owners and forest planners. When addressed, action is taken in only half of the cases, with forest professionals more frequently initiating these conversations.

Factors such as the choice of logging method can be significant, particularly in areas adjacent to settlements. In cognitive psychological research, interest is directed to holistic sensations aroused by the environment (Ulrich, 1983). The forest environment's attractiveness can be attributed to environmental variables represented by words such as

beauty, interest, surprise, diversity, harmony, forest sense, clarity, and naturalness. Clearcutting has been associated with a decrease in the recreational value of forests (Karjalainen, 2006), whereas logging methods with retention trees, such as selection cutting, are considered more socially acceptable. People prefer forests with diverse tree ages, species, and sizes, which are not too densely spaced (Tyrväinen, et al., 2017). Furthermore, people's backgrounds can influence their visual perceptions of an environment's attractiveness (Buijs et al., 2009). Complementing quantitative data with qualitative insights may provide a deeper understanding of how landscape preferences intersect with objective and subjective factors.

2.4. Location-based games

Location-based games are mobile technology-assisted games that utilise some aspect of a player's location in how they are played (Sintoris et al., 2017). They have gameplay that motivates players to explore and interact with their surroundings (Laato et al., 2023). With the growth of smartphones equipped with camera imaging and motion sensing, the number and variety of location-based games have increased. These games, such as Pokémon Go and Ingress Prime, have been used to crowdsource spatial data (Laato & Tregel, 2023).

Geocaching is an example of a globally popular location-based game where players find containers, known as geocaches, hidden by other players who share GPS coordinates. Often likened to a real-world treasure hunt, it has been ongoing for over 20 years, and there are currently more than 3 million active geocaches hidden in 191 countries (Groundspeak Inc., 2018). It has evolved into a complex information system through which players interact with their environment (Ihamäki, 2015) and includes many gamification mechanics, such as badges, levels, and leaderboards, providing numerous affordances. These features add elements of competition and fun to the experience. Gameplay generates statistics that aid the completion of challenges within the game, so players' interactions with nature are often driven by their needs in these meta-games.

Geocaching has been used in research as an active method of data collection in areas such as art (Ihamäki & Heljakka, 2017), geoscience (Lazar & Moysey, 2020), and natural resources education (Smith et al.,

2021). Passive methods have also been used to highlight the game's data as an indicator for recreational and cultural ecosystem services (Cord et al., 2015; Rosário et al., 2019). Several studies have also used geocaching in health and wellness programs to encourage physical activity among young individuals (Battista & West, 2018) and foster active ageing in older populations (Fornasini et al., 2020).

3. Materials and methods

3.1. Research sites

Four research sites were established in public and privately owned forests: two in urban areas and two in rural areas, in North Karelia (Eastern Finland) and Pirkanmaa (Western Finland) (See Fig. 1). These were chosen for their diverse landscapes (See Fig. 2) and high visitor numbers. The Rural I trail is located in the Ruunaa National Hiking Area, approximately 120 km from Joensuu, and is owned by the Finnish state. The Urban I trail is in Joensuu and has mixed ownership. The Rural II trail in Seitsemien National Park, approximately 80 km from Tampere, also contains old-growth forest. The Urban II trail is in Tampere and is owned by the municipal council. Once the locations were selected, permission to establish the trails was sought and granted from each landowner.

3.2. Geocaching trail design

A geocaching trail is a group of geocaches that are hidden along a route that is typically circular. The caches can be searched for in any order, as each is independent and does not require finding one before finding the next. However, as they are numbered (#1, #2, #3...), they are expected to be found in sequence. The trails ranged from 1.8 km to 5.7 km, containing between 7 and 11 geocaches (see Table 1).

The placement of geocaches along these trails was meticulously planned, considering factors such as trail length, the diversity of forest types along the route, and existing geocache locations. They were also carefully curated to encompass a range of forest environments, including varying ages, species compositions, and management practices.

To ensure compliance with the rules of geocaching and participation

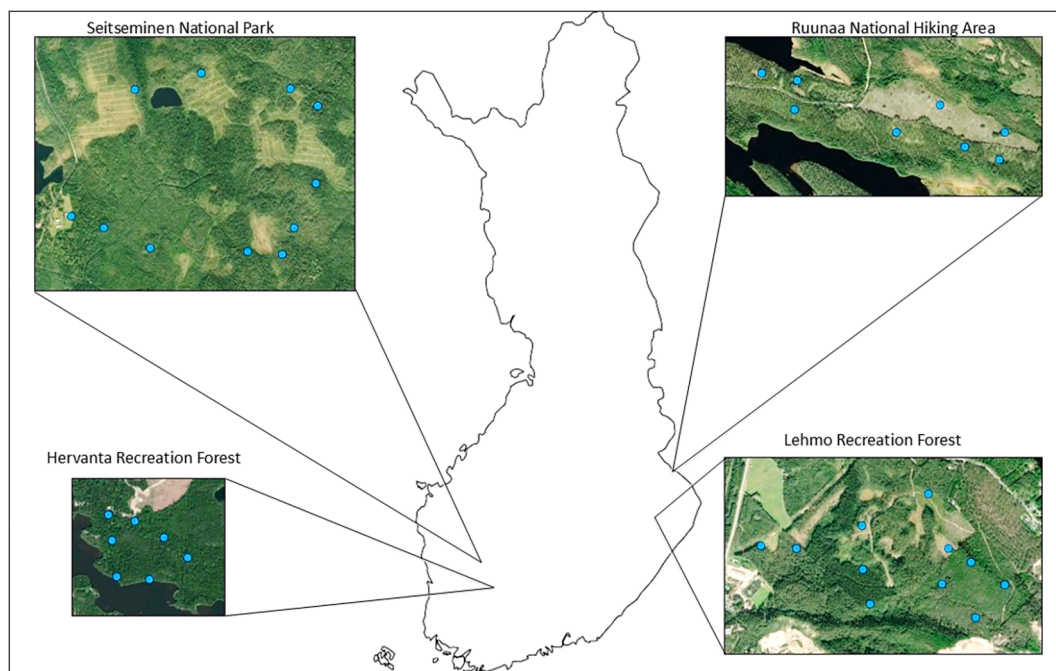


Fig. 1. Map of geocaching trail locations in Eastern and Western Finland.

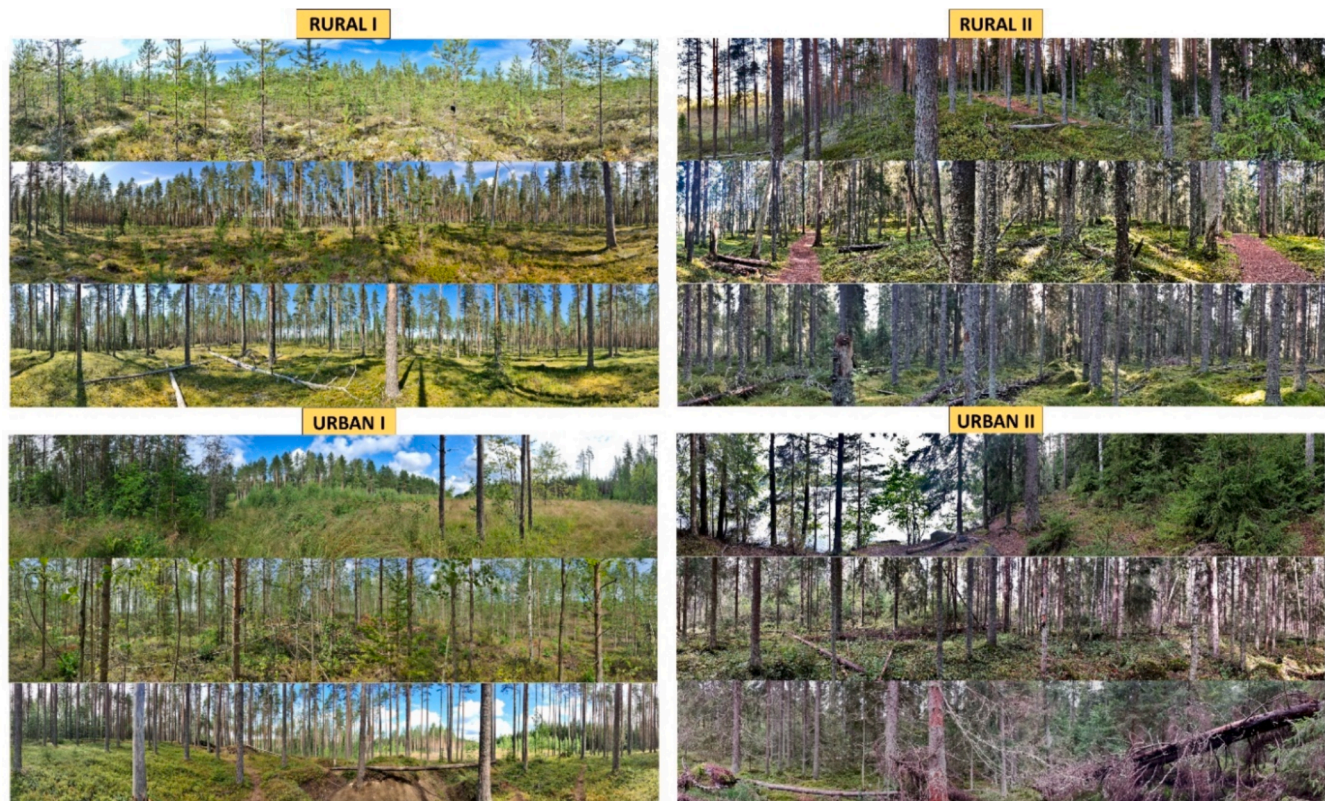


Fig. 2. Panoramic image examples from research caches across all four sites. Rural I – Ruunaa National Hiking Area, Rural II – Seitsemien National Park, Urban I – Lehmo Recreation Forest, Urban II – Hervanta Recreation Forest.

Table 1

Locations and lengths of the four geocaching trails.

Region	Trail name	Abbr.	Location	Length (km)	Number of caches
North Karelia	Urban	U1	Lehmo Recreation Forest	4.8	11
	Trail 1	R1	Ruunaa National Hiking Area	3.7	8
	Trail 1				
Pirkanmaa	Urban	U2	Hervanta Recreation Forest	1.8	7
	Trail 2	R2	Seitsemien National Park	5.7	11
	Trail 2				

in the research trails, the Finnish geocaching community were actively engaged throughout the research process through consultation with the Finnish Outdoor Association and Finnish Geocaching Association (Suomen Latu and Geonautit ry). This included organising a launch event when the Urban Trail I in Eastern Finland was opened in the summer of 2022.

The geocaching trails were set up by selecting forest areas in urban and rural settings and choosing suitable cache locations along existing path networks. Each geocache was accessible through the “Geocaching.com” mobile application, as well as open source applications like “Cachly” or “C:Geo. The webpage for each geocache included cache coordinates, descriptions in Finnish and English, difficulty and terrain ratings, a hint to find the cache, additional waypoints, a map of the cache’s location and logs and photos from previous players. Cache sites with a terrain rating of 1.5 or less were chosen to avoid the need for a Geocaching.com subscription.

3.3. Survey

In the geocache descriptions, participants were introduced to the research and informed that their participation was voluntary. The description included a link to the project’s webpages, which provided further information and details about their rights regarding the collected data. It also included a link to the survey, created using the Webropol Survey and Reporting tool, which could be opened in their mobile device’s web browser. Players were also instructed to follow Metsähallitus’ guidelines in caches placed in protected areas.

The survey asked participants questions about the forest area surrounding the geocache. Three categories of questions were created for these surveys (see Table 2).

The first category focused on human-forest relationships, primarily using qualitative, open-ended questions. Participants were prompted to reflect on how the forest made them feel and were asked which aspects of the surrounding forest pleased and displeased them. These responses, even if short, were considered as narratives reflecting the meanings of the narrator and qualitative content analysis methods were used to analyse them (Kleres, 2011). Alongside written responses, an autophotographic approach was taken, where players were asked to respond by taking photos. When in areas near active forest management or urban housing, participants were asked to what extent they agreed with statements relating to clear-cut harvesting or house construction (see Appendix A). With landscape preference questions, participants were asked about their visual preferences of the surrounding forests using numerical rating scales (e.g., 0–10), rank-ordering, and semantic differential scales.

These questions were developed from the templates (see Appendix B) to gather diverse perspectives on the specific attributes of each geocache location, enabling the exploration of attitudes towards forest features. Quantitative questions, in which participants were asked to rate the attractiveness of the surrounding landscape, were paired with

Table 2
Question categories by trail.

Region	Trail name	Location	Landscape preference questions	Human-Forest relationship questions	Photos
North Karelia	Urban Trail 1	Lehmo recreation forest	6	8	4
	Rural Trail 1	Ruunaa National Hiking Area	12	10	3
Pirkanmaa	Urban Trail 2	Hervanta Recreation Forest	6	9	2
	Rural Trail 2	Seitsemien National Park	12	13	1

qualitative questions that asked them to describe what they felt was attractive or unattractive and explain why. The quantitative questions were developed from those used in previous landscape valuation studies (Silvennoinen et al., 2022). The qualitative questions were developed specifically for this study.

3.4. Data collection

To evaluate the effectiveness of this methodology, relative response rates are analysed to determine whether the trails successfully elicit responses from the geocaching community. Next, the suitability of the survey responses for research related to human-forest relationships and landscape preferences is evaluated by analysing the responses to selected questions and ensuring they are consistent with findings from previous studies. Finally, the autophotographic submissions are evaluated qualitatively against the associated questions. Due to the exploratory nature of this study, demographic questions were asked in the first cache in each of the four geocaching trails to better understand who in the geocaching community would participate.

4. Results

4.1. Data accumulation

Data was collected between June 2022 and September 2023. Since participation in the survey was voluntary, many found the geocache but did not open the survey and participate. During this period, the total number of times a cache was found and “logged” was 3021, and the total number of valid survey responses was 966 (See the distribution between the trails in Table 3). We found that each trail’s average survey response rate ranged from 22.9 % to 38.6 %. Some participants chose to find all the caches on the trail and complete all the surveys in one day, while others visited the trail on numerous trips, completing individual cache surveys on different days. The higher English language survey response in urban areas was expected, as they tend to have more non-Finnish residents. International students and staff of nearby universities were also encouraged to participate.

4.2. Demographics of respondents

The survey in the first cache in each of the four geocaching trails included questions relating to the demographics of the survey respondents. Table 4 shows how the survey respondent type differed

Table 3
Accumulation of cache logs and survey responses across all four trails.

	Area			
	U I	R I	U II	R II
Total cache finds across the whole trail	869	343	825	984
Number of caches in trail	11	8	7	11
Average visitors per cache on the trail	79	42.9	117.9	89.5
Finnish survey respondents across the whole trail	268	97	256	208
English survey respondents across the whole trail	67	8	45	17
Combined number of respondents	335	105	301	225
Average respondents per cache	30.5	13.1	43	20.5
Average response rate	38.6	30.6	36.5	22.9

Table 4
Demographic differences across all four trails.

	North Karelia		Pirkanmaa		Total	%
	U I	R I	U I	R I		
Gender						
Male	19	7	32	15	73	49 %
Female	25	6	32	9	72	48 %
Other	0	0	0	1	1	1 %
No response	1	2	0	0	3	2 %
Age						
Under 21	0	0	1	1	2	1 %
21–30	9	0	14	0	23	15 %
31–40	9	5	12	1	27	18 %
41–50	11	1	21	5	38	26 %
51–60	7	4	10	11	32	21 %
61–70	9	4	2	5	20	13 %
Over 70	0	1	2	2	5	3 %
No response	0	2	0	0	2	1 %
Education						
None	0	1	1	0	2	1 %
School	2	1	8	2	13	9 %
College	15	6	22	9	52	35 %
University	28	7	32	13	80	54 %
Other	0	0	0	1	1	1 %
No response	0	0	1	0	1	1 %
Residence						
Countryside or sparsely populated area	2	1	5	6	14	9 %
Urban or small town (<15,000 people)	11	5	7	5	28	19 %
Medium or large city (>15,000 people)	32	9	51	14	106	71 %
No response	0	0	1	0	1	1 %
Total response	45	15	64	25	149	

between the eastern and western, and urban and rural trails. Some respondents chose not to answer these or only partially answered them.

Across all trails, the gender distribution was equal between males and females, with one respondent identifying as other, which is close to the national average of 49.50 % males to 50.50 % females (Statistics Finland, 2023). Variations were observed, such as higher male-identifying respondents in Rural Trail II in Western Finland. The combined age distribution of participants does not differ at the 5 % risk level from those at the population level. (Statistics Finland, 2023), but there was variation across the locations. The largest group (26 %) was in the 41–50 range, followed by 22 % in the 51–60 range. Age differences between the urban and rural trails were evident, with rural trails having a higher representation of older respondents than urban respondents. Respondents were educated at either college (34 %) or university level (55 %). Most respondents lived in medium or large cities (72 %).

4.3. Survey response

Fig. 3 shows the average number of times that players logged the cache as found in the geocaching.com database, survey responses per cache and the resulting response rate across all four sites. The highest number of visitors was recorded in western Finland, with 117.5 and 89.5 average visits, with the rural trail in Eastern Finland recording the

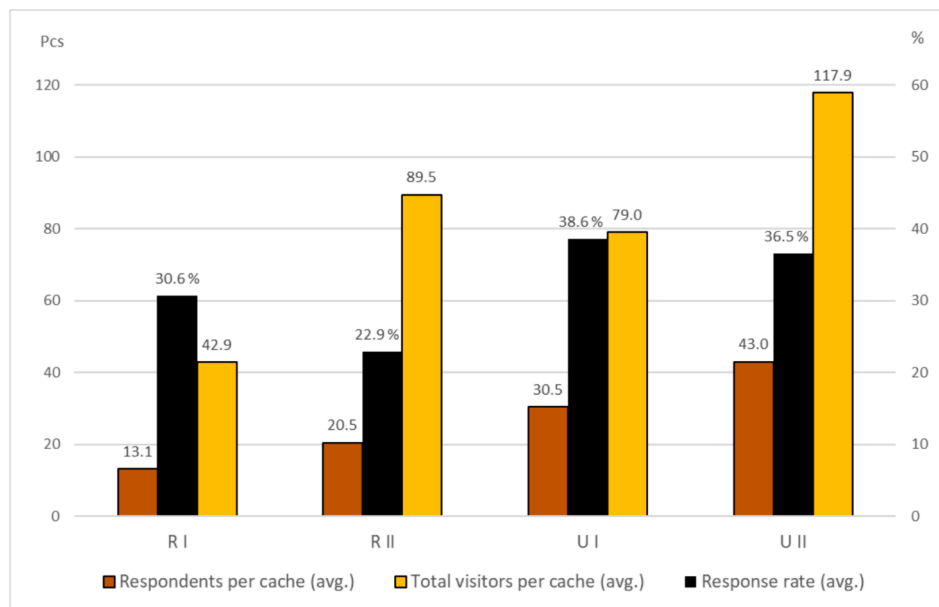


Fig. 3. Average logs, survey responses and response rate across all four sites. Rural I – Ruunaa National Hiking Area, Rural II – Seitsemien National Park, Urban I – Lehmo Recreation Forest, Urban II – Hervanta Recreation Forest.

lowest average number of visits per cache at 42.9. Furthermore, the trails in Western Finland recorded much higher numbers of visits in a much shorter time than the trails in Eastern Finland (6 months compared to 18 months), which can be seen in Fig. 5.

Fig. 4 shows how many times geocachers found each cache and how many survey responses were collected. the trendlines show a decline in response rate over each consecutive cache as participants go further along the trail. while the average number of responses per cache was relatively low (13.1–43), the response rates were moderately high (22.9 %–38.6 %). The response rate remains consistent across most locations, except for the Rural Trail II in Western Finland, where it is lower. A

higher number of responses were obtained from Urban Trail II near Tampere in just a few months compared to Urban Trail I in a year and a half, and this is likely due to Tampere's higher population density, with 1,229.2 persons/km² in 2022, compared to Joensuu, with 912.9 persons/km² (Official Statistics of Finland, 2022).

Fig. 5 shows that survey data was primarily collected during each trail's initial summer season, extending into autumn, except rural trail I. Urban areas collected more data than rural areas throughout snow-free periods. Notably, the peak response period is from mid-June to mid-July. Outdoor winter activities such as skiing and snow-shoeing are common in Finland, so while fewer responses were expected during

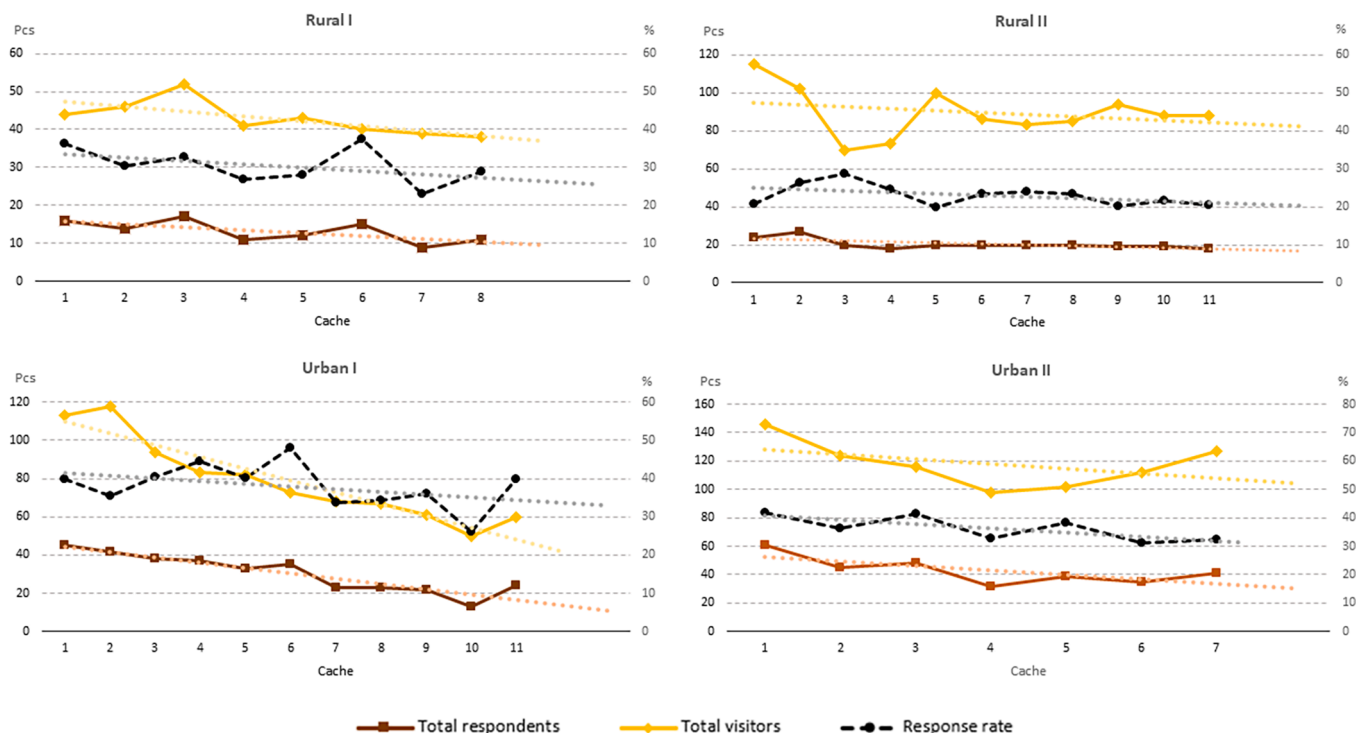


Fig. 4. Cache visits, cache responses and response rate across all four sites with trendlines.

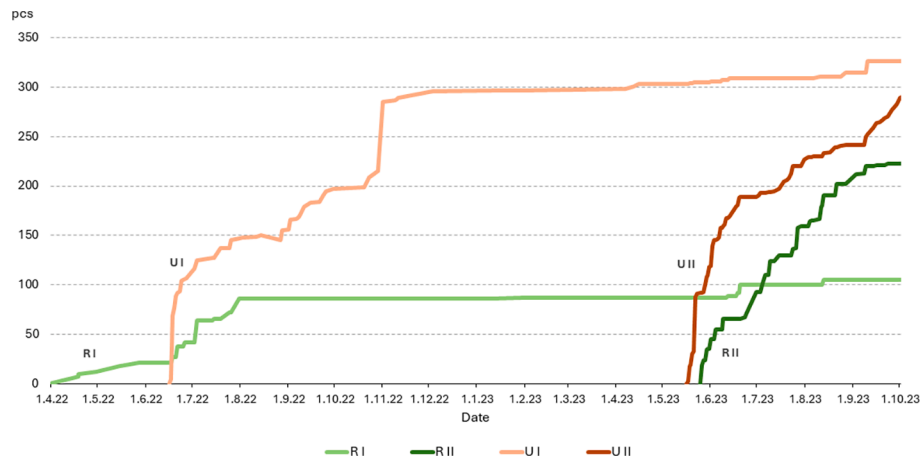


Fig. 5. Accumulation of responses across all four sites.

snow periods, the trails and surveys were kept open. In this case, survey responses only continued to accumulate on the trail closest to the largest city, Tampere.

Fig. 6 shows that the number of responses per visit varied between urban and rural areas. In urban areas, nearly one-third of the respondents visited only one cache at a time and one-fifth visited two caches simultaneously. Moreover, during their visit to caches in urban areas, respondents answered only a few cache questions. On the other hand, the number of cache responses per visit in rural areas seemed to be more evenly distributed.

In urban areas, cache visitation correlates strongly with distance from the car park. The number of responses decreases as the distance increases, with a 50 % response drop after 2 km from the car park (Fig. 7). In rural trails, there was a slight negative correlation between the total responses and the distance from the car park, indicating that in these areas, distance is not a significant factor. The first caches consistently gathered the most responses.

Qualitative and quantitative questions showed comparable response rates, while questions requiring the submission of photos had lower response rates, particularly in rural locations. Demographic questions, asked at the initial geocaches for each trail, achieved a higher response

rate than other categories (Fig. 8).

The chi-square goodness of fit tests showed significant differences in the percentages linking caches to specific respondents across the study areas (chi-square = 30.41; $df = 2$; $p < 0.001$). In rural areas, most individual cache responses correlated with those of the same person during the same visit. In contrast, this was possible for less than half of the respondents in the urban areas (see Fig. 9). Response time stamp data revealed that there were instances of multiple responses in quick succession, which suggests that geocaching in urban areas was more often a group activity than in rural areas.

Despite the experimental setup, cache logs revealed that some surveys were submitted off-site due to a misunderstanding of instructions (Fig. 10). Likely response locations were evaluated based on off-road speed (max 5 km/h = 100 m in 1.2 min), average distances between geocaches (320 m), and minimum times for finding the cache and answering the survey (total minimum of 7 min). The location could not be specified for those who had responded with only one geocache. The goodness of fit tests (chi-square = 2.683; $df = 1$; $p = 0.101$) showed no significant difference in cache response location between study areas. However, offsite responses were less common in urban areas than rural areas, although this still represented almost a quarter of the responses.

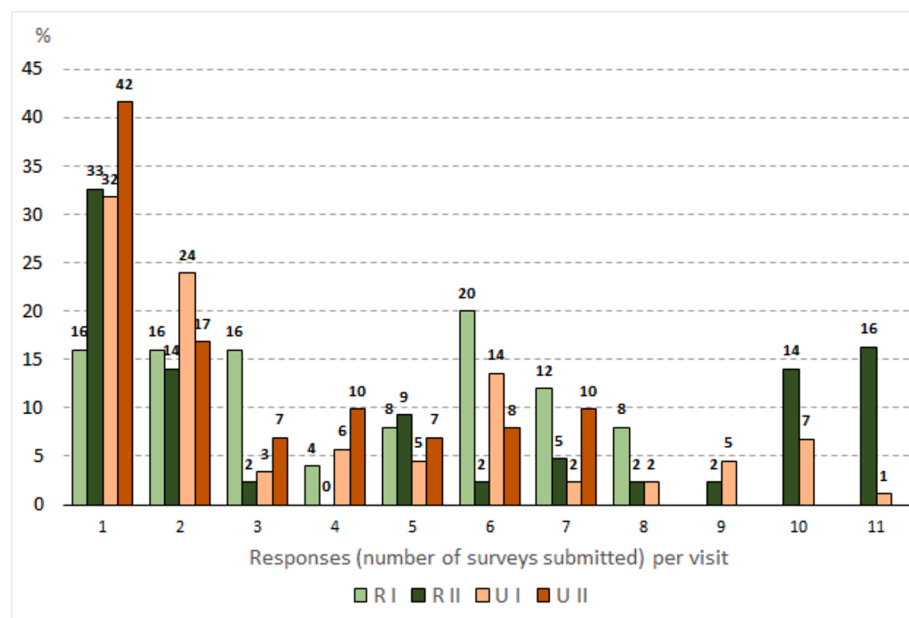


Fig. 6. Cache responses on a one-time visit across all four sites. Values reported with per cent accuracy.

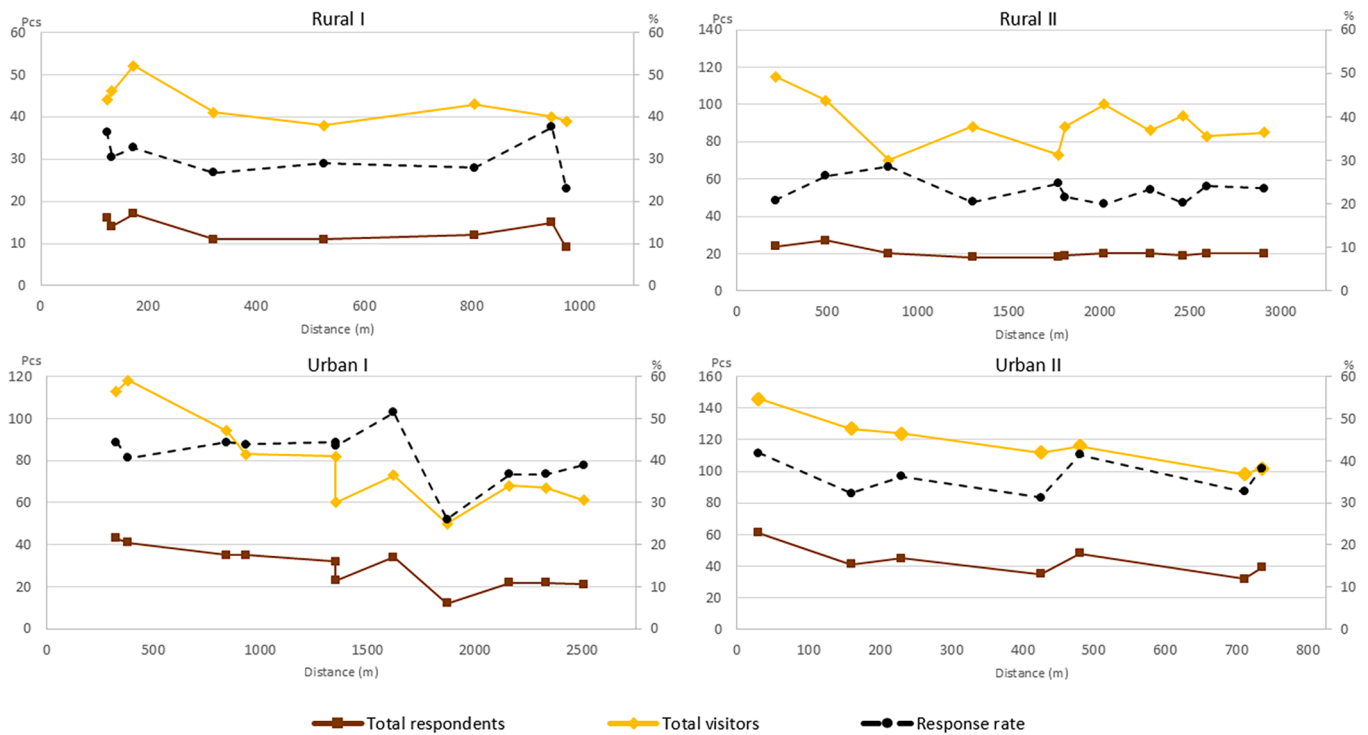


Fig. 7. The effect of the distance from the parking place to the geocaches on the cache visits, cache responses and response rate across all four sites.

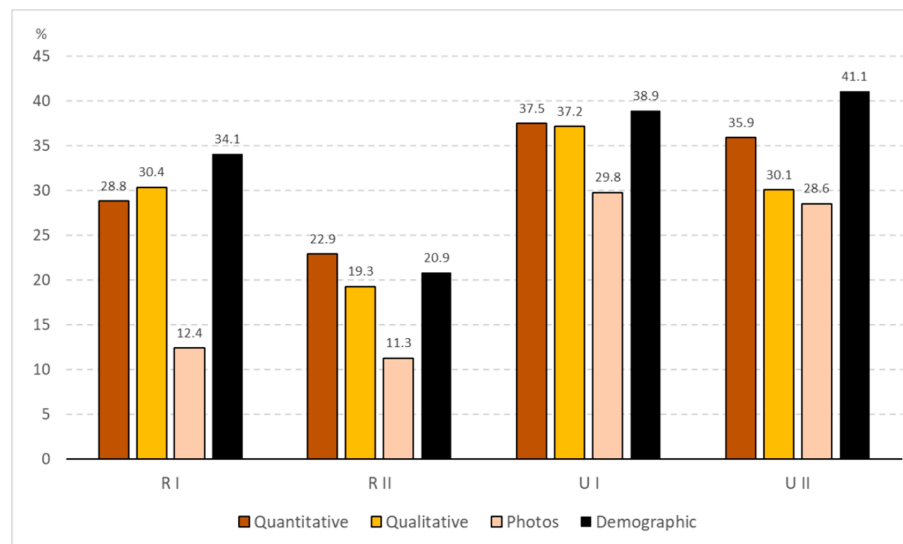


Fig. 8. Average response rate by question categories across all four sites.

4.4. Human-Forest relationships

Table 5 shows that out of the 37 caches in all four trails, 22 included open-ended questions, which yielded 591 responses. However, 63 submissions left these qualitative questions empty, resulting in a response rate of 89 % ($n = 528$). Most responses were thoughtful, related to the question, and deemed suitable for analysis. Four answers were considered unsuitable. Most of the responses were in Finnish, accounting for 454 (86 %) out of 528. English responses numbered 74 (14 %).

The length of responses varied, from only one word to several sentences, with the average being 1–2 sentences long. The analysis was conducted in Finnish for Finnish responses and in English for English responses, following an established procedure (Finfgeld-Connett, 2014).

Care was taken to consider different cultural meanings and contexts in each language. Native Finnish and English researchers conducted the study so semantic issues and meanings were verified through discussion.

The conventional content analysis involved thoroughly reading all responses for data familiarity, coding the responses by trails and languages to facilitate subsequent identification, and organisation of responses based on detectable elements such as urban and old forests, clear-cut forests, and forests with dead wood. After that, responses were sorted into 15 main themes, and short descriptions were created to summarise the characteristics of each subtheme. Finally, the interpretation of the results was grounded in the existing research literature to clarify the findings.

The 15 identified themes were *aesthetics*, *animals*, *biodiversity*, *things*

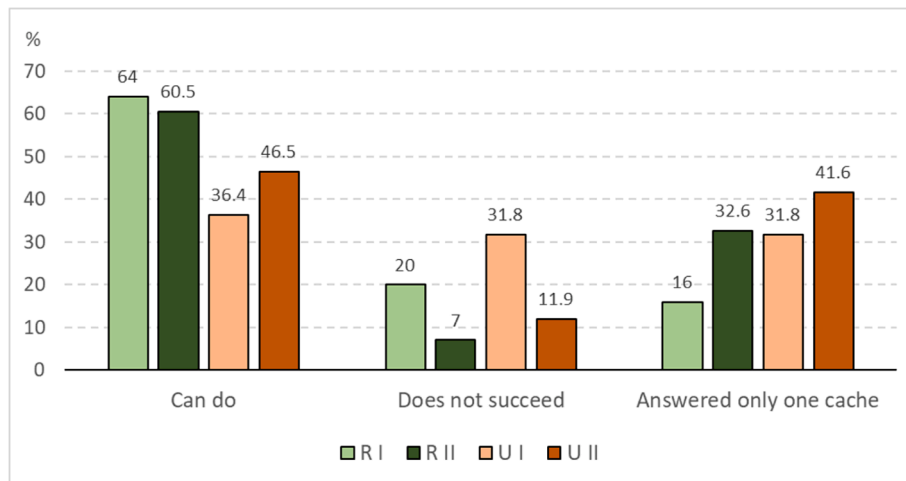


Fig. 9. Linking responses to individual participants across all four trails.

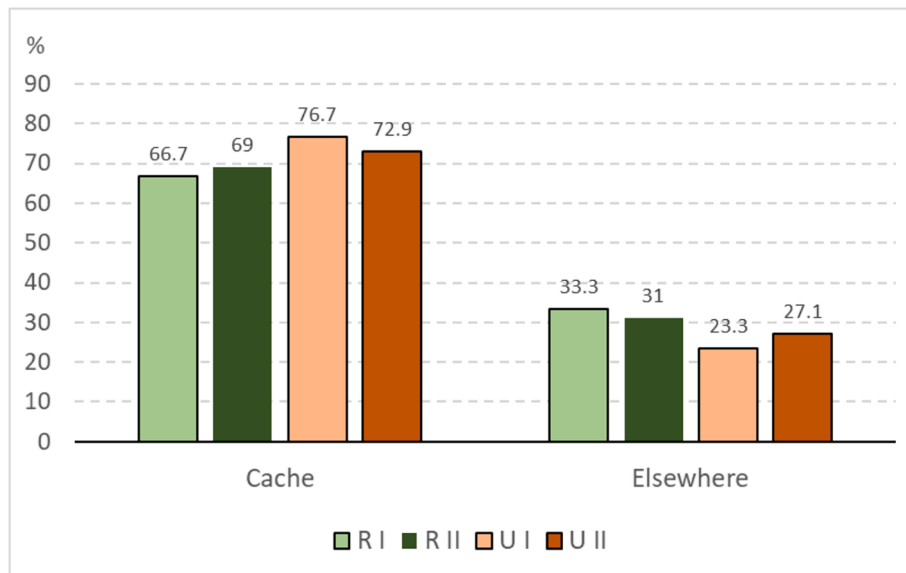


Fig. 10. Where the surveys were likely answered.

Table 5
Qualitative survey response.

Trail name	No. caches on the trail	No. of caches with qualitative questions	Total survey submissions	No. of responses to qualitative questions	Response rate (%)
Urban Trail 1	11	7	216	213	98.6
Rural Trail 1	8	4	62	58	96.5
Urban Trail 2	7	4	178	141	79.2
Rural Trail 2	11	7	135	116	85.9

that cause disgust, essence/nature of the forest, forest bounty, forest management, landscape and its details, moving in the forest, recreation and playing, the relationship between humans and nature, soundscape, time, vegetation, and wellbeing. The meanings that were identified in the responses were emotions (e.g. sorrow for clearcutting), aesthetic expressions (view, old tree), embodied and sensual experiences (human vs natural sounds, difficult terrain) or thoughts (ponderings about forest and its nature) etc.

Further analysis of the data is the subject of a future publication, however, Table 6 presents a typical example of how content analysis was conducted on a response chosen from cache number 5 on Urban Trail I.

4.5. Landscape preferences

The effectiveness of this method in eliciting responses to landscape preference questions was assessed by analysing responses from two geocache sites where participants were asked, “How well does the environment around this geocache meet your hopes and expectations for nature?”. Fig. 11 shows the results from a harvested and unharvested site on Urban Trail I in Eastern Finland. The mean score obtained from the harvested site was 4.6, notably lower than the unharvested site’s

Table 6
An example of content analysis using a response from cache number 5 on the Urban Trail I in Eastern Finland.

Question	Response
In your own words, please tell us about your thoughts and feelings about the view or detail in the cache surroundings.	Whenever I see these destroyed forests, I'm overwhelmed with sadness and gloom. And when I find something beautiful, like a wild strawberry blossom, it gives me a small hope that life will win. (L1C11)
Theme The diversity of emotions generated by clear-cutting A simultaneous experience of both beauty and horror, good and evil The aesthetics of details in the forest The experience of time	Content analysis Mourning the destruction of a forest. The comfort of a flower blooming in the soil of a destroyed forest. The hope of a flower blooming in a devastated forest that life will triumph. There can be something positive in clear-cutting: a flower blooming in an opening, giving hope for the future. A single blooming flower creates an aesthetic experience. A flower blooming in a clear-cut area gives hope for a better future, even if the surrounding forest has been destroyed.

mean score 6.9. These results align with previous studies (Arnberger et al., 2018) and validate the use of this method in collecting responses on landscape preferences.

4.6. Autophotographic responses

To evaluate the efficacy of this method in eliciting autophotographic responses, a two-phase photo-coding methodology process was followed (Byrne et al., 2022). Submissions from cache #4 on Urban Trail II in Western Finland were selected for analysis where participants were asked to “1. Take a photo of the view or detail that pleases you the most around this geocache.” and “2. Take a photo of the view or detail that pleases you the least around this geocache.” A qualitative content analysis of the collected photos was conducted. In the first phase, artefacts in the images were coded, and then, in the second phase, a thematic analysis was conducted to identify clear themes emerging from the respondents’ images.

The photos submitted for the favourite view or detail mainly featured scenes of the nearby lake and forested landscape. Those focussing on details mainly concentrated on natural elements closer to the

respondent, such as tree branches and reflections on the water. These findings sharply contrasted with the photos submitted for the least favourite view or detail. In this case, all the submitted photos depicted details near the respondent, primarily highlighting human-made objects or effects, such as litter, waste from harvesting, infrastructure, or dead parts of the surrounding forest, like deadwood and trees (see Fig. 12).

These findings illustrate that this method of creating geocaching trails with embedded surveys is suitable for collecting data for auto-photographic studies.

5. Discussion

The results demonstrate the benefits and difficulties of using surveys in geocache trails. While the results are mixed in some areas, the surveys are generally effective in gathering longitudinal place-based data for evaluating landscape preferences and human-forest relationships in urban and rural forests, which can inform forest planning decision-making processes. However, some findings could be valuable in directing future studies that use this method.

5.1. Proximity to settlements

It was found that geocaching routes closer to settlements generate better data than those located further away. This is due to higher visitor numbers and the likelihood that geocachers may find the trails more accessible, confirming previous findings (Rosário et al., 2019), which found that the most visited geocaches were the most readily available. However, the data shows that easy access may pose challenges, and some geocachers find one or two caches on multiple trips, complicating analysis. Also, geocachers tend to discover more caches in densely populated areas (Golbeck & Neustaedter, 2016), so survey trails in these areas could provide forest planners with more responses. However, it might be difficult to find suitable locations without being restricted by existing geocaches. Indeed, the full implementation of the trails in this study was delayed due to this issue, requiring multiple submissions to the geocaching reviewers. Furthermore, these results outline the potential for simultaneous data collection in both urban and rural locations, which could be helpful for future studies aiming to bridge the gap between these related lines of enquiry (Sullivan, 2022).

5.2. Optimal cache trail length

Geocaching trails with fewer caches may be more efficient for data collection than longer trails, which appear to overwhelm respondents, reflecting prior research which found that longer surveys have lower

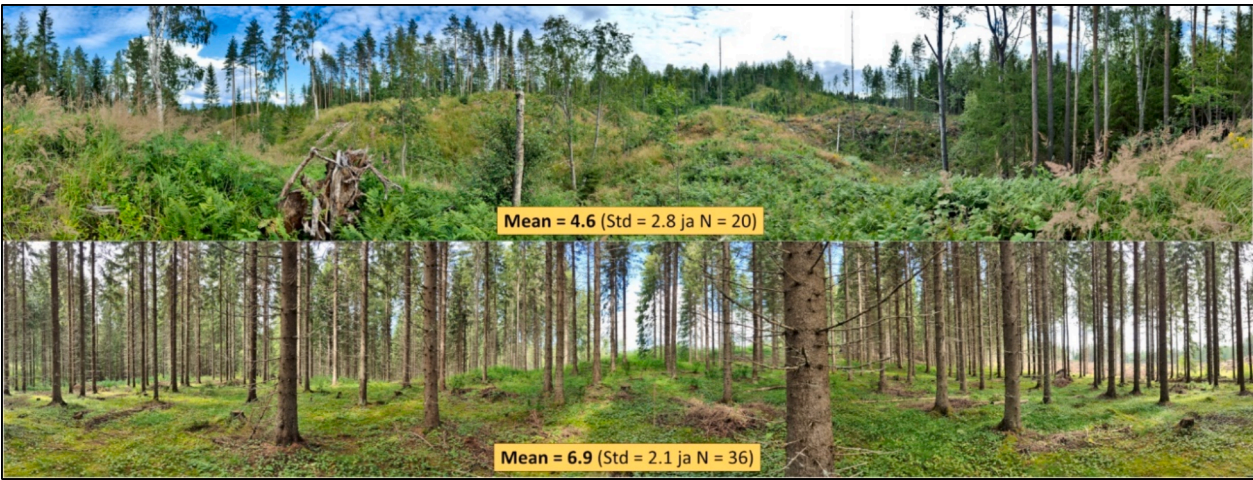


Fig. 11. Example of how answers to landscape preference questions can vary in different settings: “How well does the environment around geocache meet hopes and expectations for nature? Answer on a scale of 0–10 (0 = not at all ... 10 = completely).”

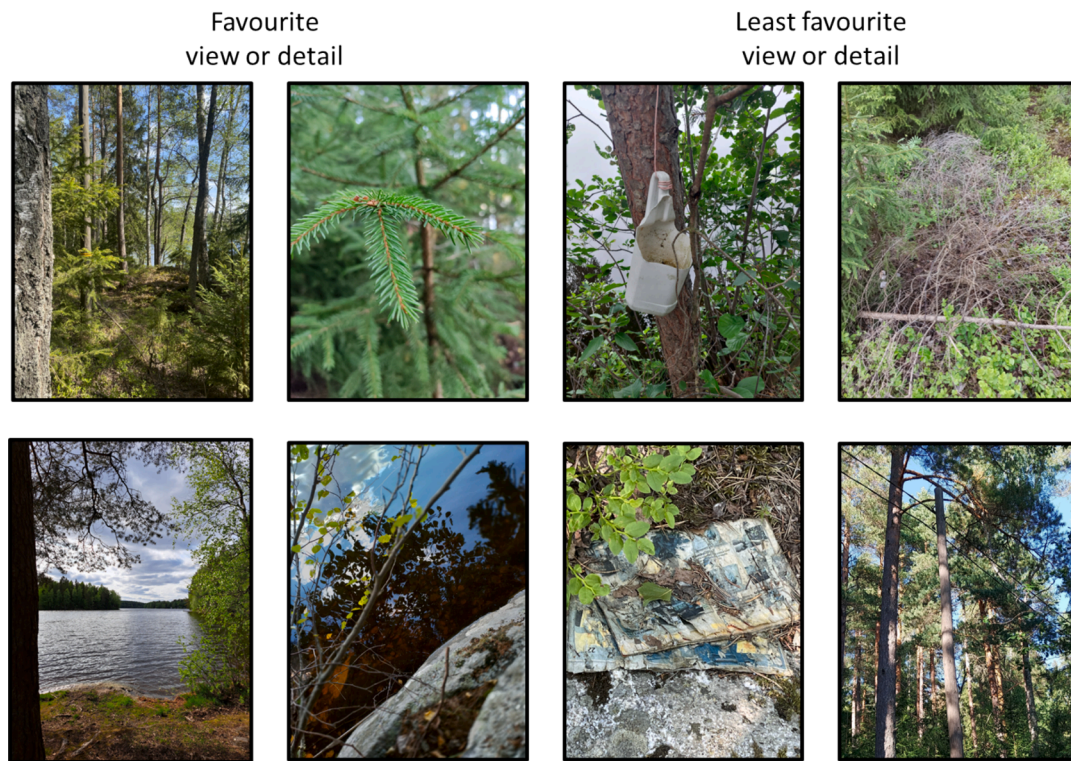


Fig. 12. Sample of photos collected of respondent's favourite and least favourite view or detail in the vicinity of cache #4 on Urban Trail II in Western Finland.

completion rates (Galesic & Bosnjak, 2009). This finding holds regardless of whether the trail is in an urban or rural setting. Breaking the route into more manageable sections, completed on separate days, was observed in the Urban I trail, while the perception of the task being too large may be a factor in the Rural II trail's response rate decline.

5.3. Location of caches

The study revealed that placing geocaches closer to the trail's starting point or car parking area enhances data gathering. This trend was consistent across all four trails, with the survey response rate decreasing as participants progressed along the route. Despite the study's intention to encourage participants to explore various forest types, the data indicates that shorter, circular routes would be more desirable. This, again, confirms findings that cache accessibility is essential for high visitation rates (Rosário et al., 2019). It is important to note, however, that geocaching imposes a minimum distance rule, preventing caches from being placed closer than 161 m (0.1 miles) from each other. Balancing establishing a route with adequate caches while keeping it close to the starting point may pose a challenge.

5.4. Cache maintenance

The data suggests that maintaining geocaches for research purposes for more than a year may not be worthwhile, as most of the data is accumulated in the first summer after the geocaching routes are established. However, some of the caches, mostly in urban areas, continued to be found during winter months, often in deep snow. Depending on the location, winter finds can be encouraged by adding cache attributes that players collect to the cache page; 'Available in winter', 'May require snowshoes', 'May require cross country skis', etc. Also, for some surveys, it may be more efficient to establish new caches annually or for a set period, although it is important to note that the Finnish Geocaching Association recommends keeping caches in place for at least six months. Caches intended for short-term use may face challenges during the review process for approval. Conversely, if trails are established to

investigate questions related to environmental change, keeping the same geocaches in operation for several years would be beneficial. In such cases, response activity can be increased by organising events in cooperation with local geocachers or other interested parties. This aligns with prior research on online survey response, which indicated that engagement with potential respondent communities beforehand resulted in higher response rates (Wu et al., 2022). Additionally, it may have the added benefit of ensuring that the caches are operational, as the community will be aware of the trail's purpose and will communicate any issues with the cache.

5.5. Demographic data

The collection of descriptive demographic data was limited to the first cache in each trail as it was judged that repeatedly answering the same questions at each cache would be detrimental. The results support this decision as the first caches of each trail, except the trail at Seitsemäminen National Park, yielded the highest number of responses. However, this made it challenging to analyse survey answers against this data, especially if respondents didn't visit the first cache. This adds weight to the recommendation that the best model for geocaching trail surveys is that they follow a clear circle loop with a clear starting point, and maybe clearer instructions may help resolve this.

The demographic questions showed that the respondents were generally highly educated, lived in more urban areas, and were older than expected, although all age groups were represented. While the youngest and oldest were underrepresented, the respondents are considered sufficiently representative of those who typically participate in forest planning processes and citizen science (Pateman et al., 2021). Furthermore, modern, multi-purpose forest planning aims to involve all forest user groups. These findings are expected to enhance understanding of the geocaching community in forest and landscape planning. It is important to note that this group of highly-motivated forest visitors may not reflect the local community of any given forest. However, in national park contexts, gathering the views of visitors from afar is common and useful.

5.6. Response rates

The results of this study indicate that embedding surveys in geocaches can effectively elicit responses to both quantitative and qualitative questions. Respondents demonstrated an equal willingness to answer both types of questions. However, there is a tendency for the response rate to decline when the survey question involves an action, such as taking photos. This finding aligns with previous research (Ilic et al., 2022), which found that, depending on the difficulty of the topic, respondents are more willing to answer survey questions than to take pictures. However, that same study highlights that pictures can contain more valuable data than written responses, so factors that may affect photo responses, such as technological issues, differing capabilities of phone models, or signal problems in rural areas, must be carefully considered.

5.7. Qualitative analysis

In the analysis of open answers relating to participants' human-forest relationships, conventional content analysis was used (Hsieh & Shannon, 2005). According to the constructivist approach, humans are regarded as active subjects construing reality by giving meanings to the world and its phenomena (Patterson, 2008). This method's primary advantage is obtaining information on participants' perceptions, thoughts, or feelings while in forest surroundings. Through various questions, participants were encouraged to describe freely how they perceived and felt in the forest without relying on preconceived categories. The answers were regarded as the meanings participants associated with the forest. In this respect, the collected responses are successful, and the methodology effectively elicits material for qualitative analysis.

5.8. Survey accessibility

Finally, by examining survey time stamps, it became evident that the survey was mistakenly completed outside the cache area. Respondents also communicated this in the geocache log or survey response. Logging at home can be common for some geocachers. Clear communication and requiring a photo to be taken at each cache location would address this.

6. Conclusion

Overall, the findings of this study underscore the potential of using location-based games to gather preference data for forest planning processes. The study had a unique approach that involved embedding surveys into geocaching trails and asking questions about human-forest relationships and landscape preferences in a real-world setting. However, there were challenges in designing and maintaining geocaching trails, survey communication and knowledge of geocaching within the research team. For further studies, involving someone with in-depth geocaching knowledge or engaging closely with the local geocaching community is recommended. Careful planning and site selection are essential for creating easily accessible trails. Survey questions should be straightforward, and a method should be incorporated to verify that submitted responses reflect experiences at the location.

Despite these challenges, survey response rates were similar to those of other data collection methods, with the added benefit of being location-specific, longitudinal, and real-time. This approach can be customised to suit the needs of landscape and forest planners in various contexts. It can be used in triangulation or combination with other active and passive methods to explore different landscape uses and perceptions. For example, it can be used in combination with PPGIS to assess opinions and feelings on conservation measures at national parks, greenspace in urban planning, or attitudes on renewable energy infrastructure in mixed-use forest plans.

This active approach to data collection introduced new concepts to

forest visitors from the geocaching community. It motivated them to participate in research through the game they love to play and may encourage them to engage with participative forest planning processes in the future. This highlights the importance of harnessing the potential of playful technologies for data collection and engaging with location-based game communities as they continue to emerge and grow.

7. Author Statement

During the preparation of this work, the author(s) used Grammarly and ChatGPT to check for grammar, spelling, and clarity. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the publication's content.

CRedit authorship contribution statement

Philip Chambers: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tuulikki Halla:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Harri Silvennoinen:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Teppo Hujala:** Writing – review & editing, Supervision, Resources. **Jukka Tikkanen:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.landurbplan.2024.105195>.

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